humidity sensor, HC-SR04 ultrasonic sensor, ACS712 current sensor, 16x2 LCD display, and several other components.

Construction and testing

To begin, upload the source code to the ESP32 board, then connect all components as shown in Fig. 4. Power the system for local data monitoring via the LCD. Fig. 5 displays local data on the LCD, while remote monitoring on the Adafruit dashboard shows real-time temperature and current data (see Fig. 6).

For transformer testing, refer to Fig. 7, which outlines various load connection diagrams. The upper diagram shows the inductive load, the middle diagram shows a single resistive load, and the lower diagram shows two resistive loads in parallel. Testing the transformer's performance involves evaluating the three distinct load conditions. The authors' prototypes are displayed in Figs 8 through 10 for the three conditions:

(a) An inductive load is applied to assess the transformer's behaviour under variable loads. The prototype is shown in Fig. 8.

(b) A single resistive load is connected. The corresponding prototype is shown in Fig. 9.

(c) Two resistive loads are connected in parallel. The prototype is shown in Fig. 10.

As the load increases in each scenario, the current drawn from the transformer also rises, while the output voltage decreases due to the transformer's internal impedance.

EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*

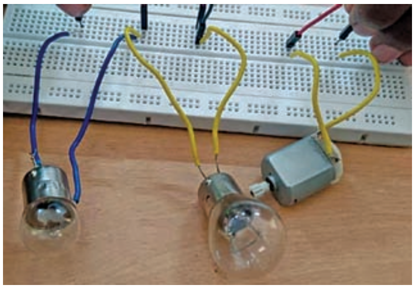

Fig. 8: Inductive load testing of the transformer

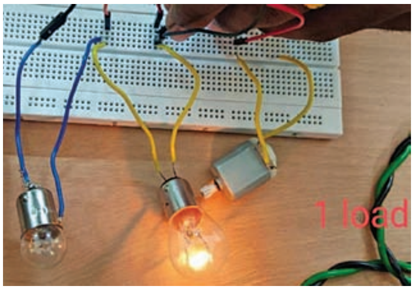

Fig. 9: Single resistive load testing of the transformer

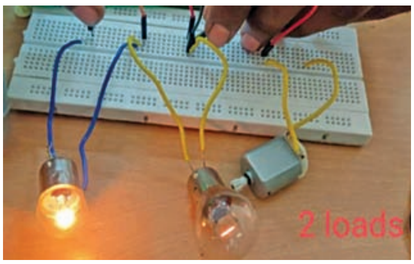

Fig. 10: Two resistive loads for testing the transformer

This behaviour is characteristic of transformers under load, where voltage regulation diminishes as the load current increases.

If the current exceeds the preset threshold of 1A, the system triggers an alert on Adafruit IO, signalling an overcurrent situation. This proactive alert helps prevent transformer damage and ensures timely interventions to maintain system reliability and safety. EFY


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